

Question 1 is compulsory.

Attempt any 3 from questions 2 to 6.

Scientific Calculator is allowed to use.

17/12/2024 FE-ALL BRANCHES SEM-I NEP2020 AM-I QP CODE: 10070769

1. Attempt any **Five** questions. (**Compulsory Problem**)
 - (a) Find all values of $(1 + i)^{1/3}$. 3
 - (b) Prove that: $\tanh(\log\sqrt{3}) = 0.5$ 3
 - (c) If $u = (1 - 2xy + y^2)^{-1/2}$, prove that $x \frac{\partial u}{\partial x} - y \frac{\partial u}{\partial y} = y^2 u^3$. 3
 - (d) Find n^{th} derivative of $y = e^{ax} \sin^2 x$. 3
 - (e) Show that the matrix $A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$ is unitary and hence find A^{-1} . 3
 - (f) Using Newton Raphson's Method, find an iterative formula for $\sqrt[3]{150}$. 3
2.
 - (a) If $H = f(y - z, z - x, x - y)$, then prove that $\frac{\partial H}{\partial x} + \frac{\partial H}{\partial y} + \frac{\partial H}{\partial z} = 0$. 4
 - (b) Prove that $\log(e^{i\alpha} + e^{i\beta}) = \log \left[2 \cos \left(\frac{\alpha-\beta}{2} \right) \right] + i \left(\frac{\alpha+\beta}{2} \right)$. 5
 - (c) Apply the Gauss Seidel method, to solve the following system of linear equations up to the two iterations. 6
 $15x + 3y - 2z = 85, x - 2y + 8z = 5, 2x + 10y + z = 51$.
3.
 - (a) Discuss the maxima and minima of $(x^2 + y^2 + 8x + 6y + 6)$. 4
 - (b) Solve the equation $x^4 + x^3 + x^2 + x + 1 = 0$. 5
 - (c) If $u = \sin^{-1} \left(\frac{x+y}{\sqrt{x}+\sqrt{y}} \right)$, then prove that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = -\frac{\sin u \cos 2u}{4 \cos^3 u}$. 6
4.
 - (a) Prove that $\operatorname{sech}^{-1}(\sin \theta) = \log \left(\cot \frac{\theta}{2} \right)$. 4
 - (b) Prove that $\cos^6 \theta = \frac{1}{32} (\cos 6\theta + 6 \cos 4\theta + 15 \cos 2\theta + 10)$. 5
 - (c) Find nonsingular matrices P & Q such that PAQ is in normal form and hence find rank of the matrix A for the following matrix $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$. 6
5.
 - (a) Find a, b, c & A^{-1} if $A = \frac{1}{9} \begin{bmatrix} 8 & -4 & a \\ 1 & 4 & b \\ 4 & 7 & c \end{bmatrix}$ is orthogonal. 4
 - (b) State & prove Euler's theorem on homogeneous functions with two independent variables. 5
 - (c) If $y = e^{\tan^{-1} x}$ then prove that $(1 + x^2)y_{n+2} + [2(n+1)x - 1]y_{n+1} + n(n+1)y_n = 0$. 6
6.
 - (a) Find n^{th} derivative of $y = \frac{x}{1+3x+2x^2}$. 4
 - (b) Investigate for what values of λ and μ the system of linear equations $x + y + z = 6; x + 2y + 3z = 10; x + 2y + \lambda z = \mu$ will have (i) No solution, (ii) Unique solution, (iii) Infinite number of solutions. 5
 - (c) Find the three sets of iterative solutions of the following equations by Gauss Jacobi method: 6
 $5x - y + z = 10; 2x + 4y = 12; x + y + 5z = -1$.

FE Sem-I (NEP) / Branches - All | Q.P. code - 99303 Nov-25
 (02-Hours) Marks-60
 Date - 22-12-25

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- Q-1 Attempt any Five questions. (Compulsory Problem) 15
- a) If $u = \frac{\sqrt{x+y}}{x+y}$ find $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$. 03
- b) If $z = \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$ and $\bar{z}$ is the conjugate of z , then prove that $(z)^{10} + (\bar{z})^{10} = 0$. 03
- c) Find n^{th} derivative of $y = \sin 2x \cos 6x$. 03
- d) Express the matrix $A = \begin{bmatrix} 2+3i & 2 & 3i \\ -2i & 0 & 1+2i \\ 4 & 2+5i & -i \end{bmatrix}$ as the sum of a Hermitian and a skew-Hermitian matrix. 03
- e) Prove that $\log \left(\frac{x+iy}{x-iy} \right) = 2i \tan^{-1} \left(\frac{y}{x} \right)$. 03
- f) Solve $10x + y + z = 12$, $x + 10y + z = 12$, $x + y + 10z = 12$ using the Gauss-Jacobi's method by taking two iterations. 03
- Q2-a) If $z = f(x, y)$, $x = r \cos \theta$, $y = r \sin \theta$ then prove that $\left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 = \left(\frac{\partial z}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial z}{\partial \theta} \right)^2$. 04
- Q2-b) Using the Newton-Raphson method find the root of the equation $2x^3 - 3x + 4 = 0$ lying between -2 and -1, correct to four places of decimals. 05
- Q2-c) If $\tan(\alpha + i\beta) = e^{i\theta}$, prove that $\alpha = \frac{n\pi + \pi}{2 + 4}$ and $\beta = \frac{1}{2} \log \left[\tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right) \right]$. 06
- Q3-a) Find the rank of the matrix by reducing it to normal form $A = \begin{bmatrix} 1 & -1 & -2 & -3 \\ 4 & 1 & 0 & 2 \\ 0 & 3 & 1 & 4 \\ 0 & 1 & 0 & 2 \end{bmatrix}$. 04
- Q3-b) For $u = \sin^{-1}(x^2 + y^2)^{1/5}$ deduce $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{2}{25} \tan u (2 \tan^2 u - 3)$. 05
- Q3-c) If $\sin^4 \theta \cos^3 \theta = a_1 \cos \theta + a_3 \cos 3\theta + a_5 \cos 5\theta + a_7 \cos 7\theta$, Then prove that $a_1 + 9a_3 + 25a_5 + 49a_7 = 0$. 06
- Q4-a) Test the consistency and hence solve the following system of linear equations, $-2y + 3z = 2$, $2x + y + z + t = -4$, $4x - 3y + z + 7t = 8$. 04
- Q4-b) Show that $\log [\sin(x + iy)] = \frac{1}{2} \log \left(\frac{\cosh 2y - \cos 2x}{2} \right) + i \tan^{-1} (\cot x \tanh y)$. 05
- Q4-c) If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, prove that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 u = -\frac{9}{(x+y+z)^2}$. 06
- Q5-a) For the matrix $A = [a_{ij}]_{3 \times 3}$, where $a_{ij} = i + j$, find the rank. 04
- Q5-b) Find all the values of $\left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right)^{3/4}$ and show that their continued product is 1. 05
- Q5-c) Discuss the maxima and minima of $x^3 y^2 (1 - x - y)$. 06
- Q6-a) Solve the following equations by Gauss-Seidel method by taking two iterations. $28x + 4y - z = 32$, $2x + 17y + 4z = 35$, $x + 3y + 10z = 24$. 04
- Q6-b) If $y = e^{m \sin^{-1} x}$, prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + m^2)y_n = 0$ 05
- Q6-c) Investigate for what values of λ and μ the system of equations $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = \mu$ will have (i) No Solution, (ii) Unique Solution (iii) Many Solutions. 06

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Applied Mathematics-I

Revised Course (NEP-2020)

(June 2025)

Marks-60

Duration-02 Hours

03/06/2025 FE (SEM-I) ALL BRANCHES (NEP-2020) AM-I QP CODE: 10091911

1. Attempt any **Five** questions. (**Compulsory Problem**)

(a) Prove that $\sqrt{1 + \operatorname{cosec}(\theta/2)} = (1 - e^{i\theta})^{-1/2} + (1 - e^{-i\theta})^{-1/2}$. 3

(b) Find n^{th} derivative of $\frac{2}{(x-1)(x-2)(x-3)}$. 3

(c) If $5 \sinh x - \cosh x = 5$, find $\tanh x$. 3

(d) Show that the matrix $A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$ is unitary and hence find A^{-1} . 3

(e) If $u = 2(ax + by)^2 - k(x^2 + y^2)$ and $a^2 + b^2 = k$, find $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$. 3

(f) Using Newton-Raphson Method, find the real root of $x^3 - 2x - 5 = 0$ correct to three decimal places. 3

2. (a) If $\tan[\log(x + iy)] = a + ib$, prove that

$$\tan[\log(x^2 + y^2)] = \frac{2a}{1 - a^2 - b^2} \text{ when } a^2 + b^2 \neq 1.$$

(b) Examine the function $f(x, y) = y^2 + 4xy + 3x^2 + x^3$ for extreme values. 5

(c) If $u = f(e^{x-y}, e^{y-z}, e^{z-x})$, then prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$. 6

3. (a) Reduce the following matrix to normal form and find the rank of matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 3 & 1 & 1 \end{bmatrix}$. 4

(b) Solve $x^6 + 1 = 0$. 5

(c) Apply Gauss-Seidel iteration method to solve the following linear equations up to the three iterations.
 $20x + y - 2z = 17, 3x + 20y - z = -18, 2x - 3y + 20z = 25$. 6

4. (a) Expand $\sin^5 \theta \cos^3 \theta$ in a series of sines of multiples of θ . 4

(b) Investigate for what values of λ and μ the equations $x + y + z = 6, x + 2y + 3z = 10, x + 2y + \lambda z = \mu$ will have (i) No Solution (ii) Unique Solution (iii) Infinitely Many Solutions. 5

(c) If $y = a \cos(\log x) + b \sin(\log x)$, prove that
 $x^2 y_{n+2} + (2n + 1)x y_{n+1} + (n^2 + 1)y_n = 0$ 6

5. (a) If $\cos \alpha + \cos \beta + \cos \gamma = 0, \sin \alpha + \sin \beta + \sin \gamma = 0$, prove that 4

i) $\cos 3\alpha + \cos 3\beta + \cos 3\gamma = 3\cos(\alpha + \beta + \gamma)$

ii) $\sin 3\alpha + \sin 3\beta + \sin 3\gamma = 3\sin(\alpha + \beta + \gamma)$

(b) If $\alpha + i\beta = \tanh\left(x + \frac{i\pi}{4}\right)$, prove that $\alpha^2 + \beta^2 = 1$. 5

(c) If $u = \sin^{-1}\left(\frac{x+y}{\sqrt{x+y}}\right)$, prove that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = -\frac{\sin u \cos 2u}{4 \cos^3 u}$. 6

6. (a) Find all the roots of $\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^{3/4}$, hence find the continued product of the roots. 4

(b) Show that the minimum value of $u = xy + a^3\left(\frac{1}{x} + \frac{1}{y}\right)$ is $3a^2$. 5

(c) Find non-singular matrices P & Q such that PAQ is in normal form & hence find rank of the matrix A and obtain A^{-1} . 6

$$A = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$